

There is a popular Web based application called Dunya, which is based on the audio processing using Essentia. This application allows the user to interact with the audio music application through music concepts related to a music culture, for example, Raaga and Taala in the case of Carnatic music. The application contains a database that includes relevant information on the music collection. The analysis module in Essentia is used to extract features from an audio recording, in order to create a meaningful relationship between all items in the database. This relationship is then utilised by the application for providing better suggestions and recommendations to the user.

Main algorithms bundled in Essentia

The Essentia package cannot be considered as a software framework. Instead, it is a useful set of algorithms for feature extraction along with some infrastructure for multi-threading and low memory usage. Some of the frequently used algorithms are:

Audio file input/output. The software comes with a variety of audio loaders and writers in order to facilitate reading and writing nearly all audio formats. An audio loader is used to load the audio file and return a stream of stereo samples. A mono loader is used to down-mix the file to a mono-signal and resample it to the given sample rate. The software also contains a large number of algorithms like AudioWriter and MonoWriter that enable writing audio files.

Standard signal processing algorithms. Essentia is bundled with major digital signal processing algorithms (DSPs) for basic processing of audio samples. The basic DSP operations of windowing, auto and cross-correlation, finding the finite Fourier transform, resampling, removing the DC component, etc can all be done using the software. We can also conveniently convert complex arrays be-

Using Essentia with Gaia

In this month's EFY Plus DVD, you will also come across a C++ package called Gaia. Both Essentia and Gaia can be used complementarily. Gaia generates classification models that could be used by Essentia to compute high-level description of music for feature extraction. This open source library comes up with Python bindings, which allow searching in a descriptor space with the help of different similarity measures and classifying results of audio analysis.

Both Essentia and Gaia are used together in a large number of research projects. These are used for the development of a search engine for music systems, automated low-level and high-level semantic description of music, audio analysis for multi-modal databases of music performances and so on.

The AcousticBrainz project

AcousticBrainz is a collaboration project between MusicBrainz community and Music Technology Group. This project aims to create a large-scale music database and make content open to the public. A project of this kind will facilitate researchers for a lot of innovations on music technology as well as open source hackers to come up with interesting recommendation engines.

The power of Essentia in music analysis is utilised in the preparation of database. The information in the database includes acoustic characteristics of music such as low-level spectral information, rhythm, keys and scales, as well as the required information for information for genres, moods and instrumentation.

tween polar and Cartesian coordinate systems using this package.

Filters. A good variety of audio-filtering operations can be carried out using Essentia. It includes low-pass, high-pass, band-pass and band-reject filtering operations. We could conveniently apply an equal loudness curve-approximating filter for filtering operations. Filtering out the DC component from the signal is also possible.

Statistics. When we are dealing with a huge amount of data, it becomes necessary to compute its statistics. It is possible to find out the energy of an array of values, as well as the root-mean-square value of these. We can have the mean, median, geometric mean and power mean of an array of values. In order to aid computation of probability distributions, variance, skewness and kurtosis algorithms are present. We can also have a single Gaussian estimate of the given list of arrays.

MIR descriptors. When it comes to music information retrieval (MIR), there is a good number of algorithms to describe the pitch, tone, rhythm, time-domain description, low-level spectrum description, middle- and

high-level descriptions and SFX descriptions for short sounds, among others.

Extractors. There are composite algorithms used in Essentia where one algorithm can contain multiple algorithms. A few useful extractors, commonly used in the retrieval of music information, have been written as algorithms. We have LevelExtractor for computation of loudness of an audio track, tuning frequency extractor, low-level spectral extractor and many more.

A versatile audio-processing library

The robustness, performance and optimality of audio-processing algorithms bundled up in Essentia make it truly versatile and ideal for research as well as industrial applications. The software is easily expandable and can be used for adapting to the requirements of particular applications. The software is truly cross-platform, supporting Linux, Mac OS X and Windows, and is optimised in terms of the computational cost of algorithms. ●

The author is an electronics enthusiast